

**INTRAVAGINAL LONGITUDINAL WEARABLE — MECHANICAL ARCHITECTURE,
ELECTRODE CONFIGURATIONS, POWER, AND STANDALONE USE CASES
(NON-LIMITING)**

A. Mechanical Architecture and Stable Intravaginal Coupling (Non-Limiting)

In various embodiments, the intravaginal longitudinal wearable comprises a sealed body containing electronics, a power source, and one or more sensing regions arranged to maintain stable tissue-adjacent coupling during movement and prolonged wear. Non-limiting structural approaches include: 1. a soft outer shell (e.g., silicone or other biocompatible elastomer) formed over an internal frame; 2. a contoured profile configured to resist migration and maintain repeatable orientation; 3. one or more compressible regions configured to maintain contact integrity across posture changes; and 4. distributed sensing regions arranged along the body to support multi-zone measurements. In certain embodiments, the device includes a retrieval feature (e.g., flexible tab or loop) and a geometry configured to reduce snagging and improve comfort.

B. Electrode Configurations for Bio-Impedance and Dielectric Sensing (Non-Limiting)

In various embodiments, bio-impedance sensing is performed using one or more electrode configurations including, without limitation: 1. two-electrode configurations for simplified measurement; 2. four-electrode configurations to reduce electrode-interface sensitivity; 3. multi-electrode arrays enabling multi-zone impedance profiles; and 4. multi-frequency excitation enabling frequency-dependent feature extraction. In certain embodiments, capacitive wetness/dielectric sensing is performed using capacitive electrodes and/or dielectric sensing regions arranged to estimate wetness trends and tissue-adjacent dielectric changes. In certain embodiments, impedance electrodes and capacitive electrodes are physically separated and/or electronically time-multiplexed to reduce interference.

C. pH Sensing Implementations (Non-Limiting)

In various embodiments, pH sensing comprises one or more solid-state pH sensing implementations including, without limitation: 1. ISFET-based pH sensing; 2. solid-state pH sensors integrated into a sealed sensing region; and/or 3. pH sensing regions configured to reduce drift and contamination. In certain embodiments, the system includes stabilization routines and drift modeling for pH outputs and gates reporting based on pH sensor quality indices.

D. Pressure and Pelvic Activity Sensing Implementations (Non-Limiting)

In various embodiments, pelvic activity and tissue interaction are captured using one or more pressure sensing approaches including: 1. distributed pressure sensors arranged in multiple zones; 2. capacitive pressure sensing; 3. piezoresistive sensors and/or force sensing resistors; and/or 4. MEMS-based sensing elements. In certain embodiments, the system derives activity indices from pressure dynamics and may differentiate sustained contact changes from transient events using IMU-informed gating.

E. Power, Charging, and Wear Detection (Non-Limiting)

In various embodiments, the device includes a rechargeable battery and power management circuitry configured for multi-hour wear sessions. Non-limiting features include: 1. duty-cycled sensing to extend runtime; 2. charging via docking base; 3. contact-based or inductive charging; and 4. battery health

monitoring and charge-state estimation. In certain embodiments, wear detection is performed using one or more indicators including thermal coupling, capacitive proximity/contact confirmation, and/or impedance plausibility.

F. Standalone Use Cases (Non-Limiting)

In standalone embodiments, Node 2 provides privacy-preserving longitudinal outputs for a single user without requiring other nodes. Non-limiting standalone use cases include: 1. generating multi-hour trend curves for moisture/dielectric indices and impedance-derived tissue state indices; 2. generating pH trend indices with confidence labeling and drift-aware normalization; 3. generating pelvic activity indices with motion gating and contact integrity scoring; 4. producing stability/variability summaries across repeated wear sessions; and 5. generating confidence-weighted summaries that exclude low-quality windows.

G. Optional Integration Hooks and Variations (Non-Limiting)

In optional multi-node embodiments, derived outputs may be time-aligned and correlated with other nodes under consent gating and quality thresholds. Any embodiment described herein may omit sensors, add sensors, substitute sensor modalities with equivalents, alter electrode layouts, alter housing geometry, alter duty cycles, and/or alter processing distribution (on-device, on-node, cloud) while maintaining the functional intent of intravaginal longitudinal capture, validity gating, and privacy-preserving outputs.